

Differential Equations and Linear Algebra

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All errors, typographical and substantive, and other offenses, are entirely my own.

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8 - Fourier and Laplace Transforms

8.1 - Fourier Series

Question: 8.1.1

(a) To prove that $\cos(nx)$ is orthogonal to $\cos(kx)$ when $k \neq n$, use the formula $(\cos(nx))(\cos(kx)) = \frac{1}{2} \cos((n+k)x) + \frac{1}{2} \cos((n-k)x)$. Integrate from $x = 0$ to $x = \pi$. What is $\int \cos^2(kx) dx$?

(b) From 0 to π , $\cos(x)$ is **not** orthogonal to $\sin(x)$. The period has to be 2π :

Find $\int_0^\pi \sin(x) \cos(x) dx$ and $\int_{-\pi}^\pi \sin(x) \cos(x) dx$ and

$$\int_0^{2\pi} \sin(x) \cos(x) dx.$$

(a) For $k \neq n$, integrate

$$\begin{aligned} \int_0^\pi \cos(nx) \cos(kx) dx &= \int_0^\pi \left[\frac{1}{2} \cos((n+k)x) + \frac{1}{2} \cos((n-k)x) \right] dx \\ &= \frac{1}{2} \left[\frac{1}{n+k} \sin((n+k)x) + \frac{1}{n-k} \sin((n-k)x) \right] \Big|_0^\pi \\ &= 0 \end{aligned}$$

When equality holds, we have

$$\int_0^\pi \cos^2(kx) dx = \int_0^\pi \left[\frac{1}{2} \cos(2kx) + \frac{1}{2} \right] dx = \frac{\pi}{2}$$

(b) I think the textbook has a typo. The integrand should be $\sin(2x) \cos(x)$.

$$\begin{aligned} \int_0^\pi \sin(2x) \cos(x) dx &= \int_0^\pi 2 \sin(x) \cos^2(x) dx \\ &= -\frac{2}{3} \cos^3(x) \Big|_0^\pi \\ &= \frac{4}{3} \end{aligned}$$

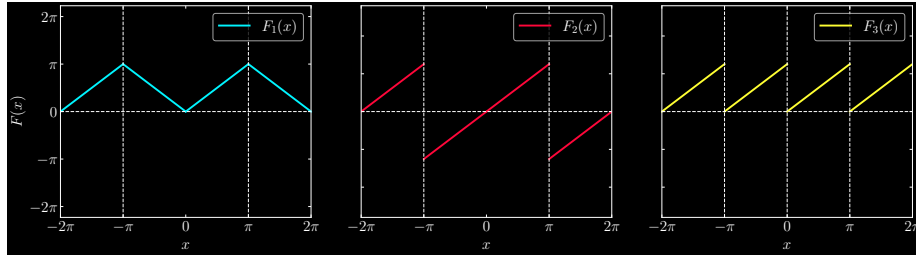
When we set the integral to the two other sets of bounds, we find

$$-\frac{2}{3} \cos^3(x) \Big|_{-\pi}^\pi = -\frac{2}{3} \cos^3(x) \Big|_0^{2\pi} = 0$$

Question: 8.1.2

Suppose $F(x) = x$ for $0 \leq x \leq \pi$. Draw graphs for $-2\pi \leq x \leq 2\pi$ to show three extensions of F : a 2π -periodic even function and a 2π -periodic odd function and a π -periodic function.

Graph each extension F_1 , F_2 , and F_3 as

**Question: 8.1.3**

Find the Fourier series on $-\pi \leq x \leq \pi$ for

- (a) $f_1(x) = \sin^3(x)$, an odd function (sine series, only two terms)
- (b) $f_2(x) = |\sin(x)|$, an even function (cosine series)
- (c) $f_3(x) = x$ for $-\pi \leq x \leq \pi$ (sine series with jump at $x = \pi$)

- (a) The Fourier coefficients are given by

$$b_k = \frac{2}{\pi} \int_{-\pi}^{\pi} \sin^3(x) \sin(kx) dx$$

The simplest approach is to write the identity $\sin(x) = (e^{ix} - e^{-ix})/2i$. Then we can calculate

$$\begin{aligned} \sin^3(x) &= -\frac{1}{8i} (e^{ix} - e^{-ix})^3 \\ &= -\frac{1}{8i} (e^{3ix} - e^{-3ix} - 3(e^{ix} - e^{-ix})) \\ &= -\frac{1}{4} (\sin(3x) - 3\sin(x)) \end{aligned}$$

which is a trigonometric identity that itself is also the Fourier series for $\sin^3(x)$, and therefore $b_1 = 3/4$ and $b_2 = -1/4$.

- (b) Note that the absolute value means

$$|\sin(x)| = \begin{cases} \sin(x) & 0 \leq x \leq \pi \\ -\sin(x) & -\pi \leq x < 0 \end{cases}$$

So calculating a_0 yields

$$a_0 = \frac{1}{2\pi} \left[\int_0^{\pi} \sin(x) dx + \int_{-\pi}^0 (-\sin(x)) dx \right] = \frac{2}{\pi}$$

Deriving the Fourier coefficients, we find

$$\begin{aligned} a_k &= \frac{2}{\pi} \int_0^\pi \sin(x) \cos(kx) \, dx \\ &= -\frac{1}{\pi} \left[\frac{\cos(1-k)x}{1-k} + \frac{\cos(1+k)x}{1+k} \right] \Big|_0^\pi \\ &= -\frac{1}{\pi(k^2-1)} [(1+k) \cos(1-k)x + (1-k) \cos(1+k)x] \Big|_0^\pi \end{aligned}$$

Now, when k is odd, this expression vanishes. Otherwise, for even k it reduces to

$$\begin{aligned} &= \frac{1}{\pi(k^2-1)} [-2(1+k) - 2(1-k)] \\ &= \frac{4}{\pi(1-k^2)} \end{aligned}$$

To cleanly put this in a summation, let $k = 2n$. Then

$$|\sin(x)| = \frac{2}{\pi} + \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{\cos(2nx)}{1-4n^2}, \quad -\pi \leq x \leq \pi$$

(c) The Fourier coefficients are

$$\begin{aligned} b_k &= \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin(kx) \, dx \\ &= \frac{1}{\pi} \left[\frac{\sin(kx)}{k^2} - \frac{x \cos(kx)}{k} \right] \Big|_{-\pi}^{\pi} \end{aligned}$$

which is $2/k$ when k is odd and $-2/k$ when k is even. By symmetry, $b_0 = 0$. The Fourier sine series is

$$x = 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin(nx)$$

Question: 8.1.4

Find the complex Fourier series $e^x = \sum c_k e^{ikx}$ on the interval $-\pi \leq x \leq \pi$. The even part of a function is $\frac{1}{2}(f(x) + f(-x))$, so that $f_{\text{even}}(x) = f_{\text{even}}(-x)$. Find the cosine series for f_{even} and the sine series for f_{odd} . Notice the jump at $x = \pi$.

First, we calculate the coefficients to the complex Fourier series as

$$\begin{aligned} c_k &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^x e^{-ikx} \, dx \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{(1-ik)x} \, dx \\ &= \frac{1}{2\pi} \frac{e^{(1-ik)x}}{1-ik} \Big|_{-\pi}^{\pi} \\ &= \frac{1}{2\pi(1-ik)} [e^{(1-ik)\pi} - e^{-(1-ik)\pi}] \end{aligned}$$

Next, decompose the exponential function into the well-known identity

$$e^x = \underbrace{\frac{1}{2}(e^x + e^{-x})}_{\text{even}} + \underbrace{\frac{1}{2}(e^x - e^{-x})}_{\text{odd}} = \cosh(x) + \sinh(x)$$

The Fourier cosine series for the even component is

$$\begin{aligned} a_k &= \int_{-\pi}^{\pi} \frac{1}{2}(e^x + e^{-x}) \cos(kx) \, dx \\ &= \frac{2}{\pi} \int_0^{\pi} \frac{1}{2}(e^x + e^{-x}) \cos(kx) \, dx \\ &= \frac{1}{\pi} \left[\frac{e^x(\cos(kx) + k \sin(kx))}{1+k^2} + \frac{e^{-x}(-\cos(kx) + k \sin(kx))}{1+k^2} \right] \Big|_0^{\pi} \\ &= \frac{(-1)^k}{\pi} \left(\frac{e^{\pi} - e^{-\pi}}{1+k^2} \right) \\ &= (-1)^k \frac{2}{\pi(1+k^2)} \sinh(\pi) \end{aligned}$$

And for the Fourier sine series for the odd component, we have

$$\begin{aligned} b_k &= \frac{1}{\pi} \int_{-\pi}^{\pi} \frac{1}{2}(e^x - e^{-x}) \sin(kx) \, dx \\ &= \frac{2}{\pi} \int_0^{\pi} \frac{1}{2}(e^x - e^{-x}) \sin(kx) \, dx \\ &= \frac{1}{\pi} \left[\frac{e^x(\sin(kx) - k \cos(kx))}{1+k^2} - \frac{e^{-x}(-\sin(kx) - k \cos(kx))}{1+k^2} \right] \Big|_0^{\pi} \\ &= (-1)^{k+1} \frac{k}{\pi} \left(\frac{e^{\pi} - e^{-\pi}}{1+k^2} \right) \\ &= (-1)^{k+1} \frac{2k}{\pi(1+k^2)} \sinh(\pi) \end{aligned}$$

Additionally, the average is

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^x \, dx = \frac{1}{2\pi} (e^{\pi} - e^{-\pi}) = \frac{1}{\pi} \sinh(\pi)$$

The full Fourier series for our $F(x)$ is then

$$e^x = \frac{1}{\pi} \sinh(\pi) + \sum_{k=1}^{\infty} (-1)^k \frac{2}{\pi(1+k^2)} \sinh(\pi) \cos(kx) - \sum_{k=1}^{\infty} (-1)^k \frac{2k}{\pi(1+k^2)} \sinh(\pi) \sin(kx)$$

Question: 8.1.5

From the energy formula (21), the square wave sine coefficients satisfy

$$\pi(b_1^2 + b_2^2 + \cdots) = \int_{-\pi}^{\pi} |SW(x)|^2 dx = \int_{-\pi}^{\pi} 1 dx = 2\pi$$

Substitute the numbers b_k from equation (8) to find that $\pi^2 = 8\left(1 + \frac{1}{9} + \frac{1}{25} + \cdots\right)$.

The coefficients b_k are

$$b_k = \frac{2}{\pi} \left\{ \frac{2}{1}, \frac{2}{3}, \frac{2}{5}, \cdots \right\}$$

Substituting gives us

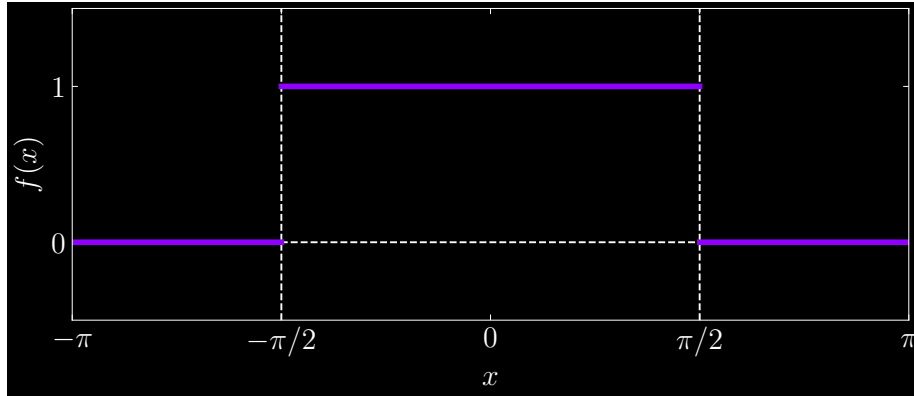
$$16\pi \left(\frac{1}{\pi^2} + \frac{1}{9\pi^2} + \frac{1}{25\pi^2} + \cdots \right) = 2\pi \quad \Rightarrow \quad \pi^2 = 8 \left(1 + \frac{1}{9} + \frac{1}{25} + \cdots \right)$$

Question: 8.1.6

If a square pulse is centered at $x = 0$ to give

$$f(x) = 1 \quad \text{for } |x| < \frac{\pi}{2}, \quad f(x) = 0 \quad \text{for } \frac{\pi}{2} < |x| < \pi,$$

draw its graph and find its Fourier coefficients a_k and b_k .



On visual inspection, it is apparent that $f(x)$ is an even function. From that we can immediately conclude that all $b_k = 0$. Then we calculate

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} dx = \frac{1}{2}$$

$$\begin{aligned}
 a_k &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(kx) \, dx \\
 &= \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} \cos(kx) \, dx \\
 &= \frac{2}{k\pi} \sin\left(\frac{k\pi}{2}\right)
 \end{aligned}$$

giving us complete Fourier series

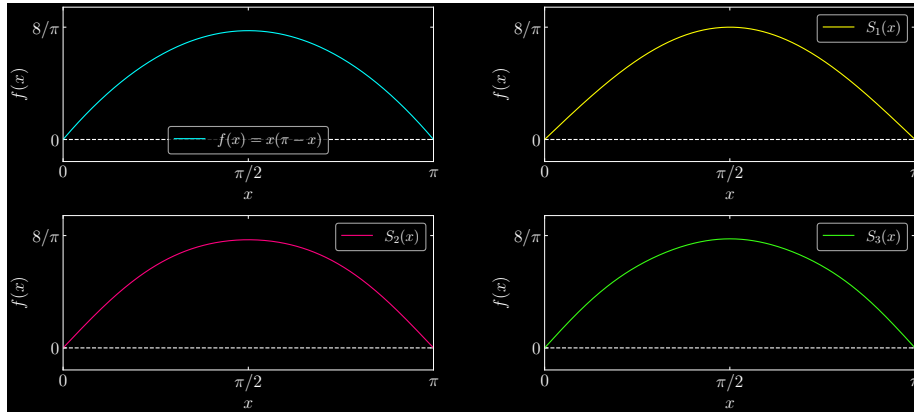
$$f(x) = \frac{1}{2} + \sum_{k=1}^{\infty} \frac{2}{k\pi} \sin\left(\frac{k\pi}{2}\right) \cos(kx)$$

Question: 8.1.7

Plot the first three partial sums and the function $x(\pi - x)$:

$$x(\pi - x) = \frac{8}{\pi} \left(\frac{\sin(x)}{1} + \frac{\sin(3x)}{27} + \frac{\sin(5x)}{125} + \cdots \right), \quad 0 < x < \pi$$

Why is $1/k^3$ the decay rate for this function? What is its second derivative?



The second derivative is $f'' = -2$. One analysis here is we must integrate thrice to derive our Fourier coefficient: twice for the x^2 term, and once more for the sine. This generates $1/k^3$.

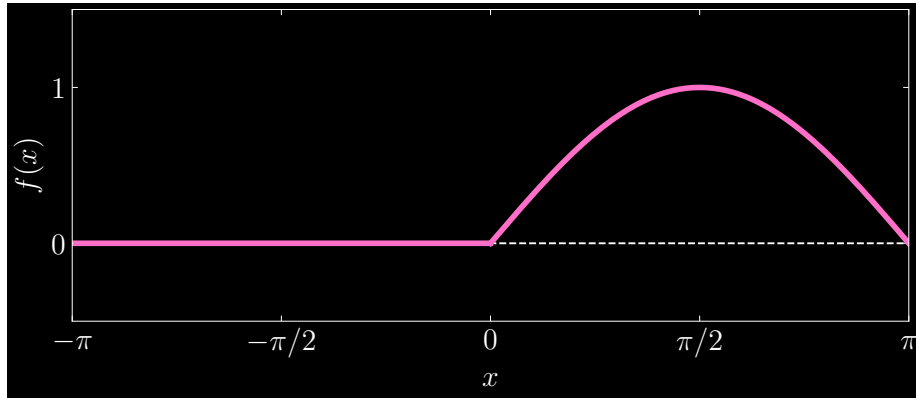
Another way to think about the decay rate is as follows: $1/k$ corresponds to jumps in the function itself (step functions), $1/k^2$ is the jump in the first derivative of the ramp function, and $1/k^3$ for the jump in the second derivative. The argument here, then, requires us to consider the periodic odd extension of f , namely the case where

$$f = x\pi + x^2, \quad -\pi < x < 0$$

In this “branch”, the second derivative is $+2$. At $x = 0$, we have a discontinuity in the second derivative: the jump from $+2$ to -2 .

Question: 8.1.8

Sketch the 2π -periodic half wave with $f(x) = \sin(x)$ for $0 < x < \pi$ and $f(x) = 0$ for $-\pi < x < 0$. Find its Fourier series.



The Fourier series is calculated as

$$\begin{aligned}
 a_0 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \, dx \\
 &= \frac{1}{2\pi} \int_0^{\pi} \sin(x) \, dx \\
 &= \frac{1}{\pi} \\
 a_k &= \frac{1}{\pi} \int_0^{\pi} \sin(x) \cos(kx) \, dx \\
 &= \frac{1}{\pi} \left[\frac{\cos((1-k)x)}{2(1-k)} - \frac{\cos((1+k)x)}{2(1+k)} \right] \Big|_0^{\pi} \\
 &= \frac{1}{\pi} \left[\frac{\cos(k\pi) + 1}{2(1-k)} + \frac{\cos(k\pi) + 1}{2(1+k)} \right] \\
 \Rightarrow a_n &= \frac{1}{\pi} \left[\frac{1}{1-2n} + \frac{1}{1+2n} \right] \quad (\text{substitute } k = 2n; \text{ non-zero only for even } k) \\
 &= \frac{2}{\pi(1-4n^2)} \\
 b_k &= \frac{1}{\pi} \int_0^{\pi} \sin(x) \sin(kx) \, dx \\
 &= \frac{1}{\pi} \left[\frac{x}{2} - \frac{\sin(2x)}{4} \right] \Big|_0^{\pi} \\
 &= \frac{1}{2}
 \end{aligned}$$

which gives us

$$f(x) = \frac{1}{\pi} + \sum_{n=1}^{\infty} \frac{2}{\pi(1-4n^2)} + \frac{\sin(x)}{2}$$

Question: 8.1.9

Suppose $G(x)$ has period $2L$ instead of 2π . Then $G(x + 2L) = G(x)$. Integrals go from $-L$ to L or from 0 to $2L$. The Fourier formulas change by a factor π/L :

The coefficients in

$$G(x) = \sum_{-\infty}^{\infty} C_k e^{ik\pi x/L} \text{ are } C_k = \frac{1}{2L} \int_{-L}^L G(x) e^{-ik\pi x/L} dx.$$

Derive this formula for C_k : Multiply the first equation for $G(x)$ by _____ and integrate both sides. Why is the integral on the right side equal to $2LC_k$?

Multiply the first equation by $e^{-ik\pi x/L}$.

$$G(x) e^{-ik\pi x/L} = e^{-ik\pi x/L} \sum_{-\infty}^{\infty} C_k e^{ik\pi x/L}$$

By orthogonality of the complex exponentials, after integrating, we are left only with

$$\int_{-L}^L G(x) e^{-ik\pi x/L} dx = \int_{-L}^L C_k dx = 2LC_k$$

Question: 8.1.10

For G_{even} , use Problem 9 to find the cosine coefficient A_k from $(C_k + C_{-k})/2$:

$$G_{\text{even}}(x) = \sum_0^{\infty} A_k \cos\left(\frac{k\pi x}{L}\right) \text{ has } A_k = \frac{1}{L} \int_0^L G_{\text{even}}(x) \cos\left(\frac{k\pi x}{L}\right) dx.$$

G_{even} is $\frac{1}{2}(G(x) + G(-x))$. Exception for $A_0 = C_0$: Divide by $2L$ instead of L .

Calculate

$$\begin{aligned} \frac{c_k + c_{-k}}{2} &= \frac{1}{4L} \left[\int_{-L}^L G(x) e^{-ik\pi x/L} dx + \int_{-L}^L G(x) e^{ik\pi x/L} dx \right] \\ &= \frac{1}{2L} \int_{-L}^L G(x) \cos\left(\frac{k\pi x}{L}\right) dx \\ &= \frac{1}{2L} \int_{-L}^L \left[\underbrace{\frac{1}{2}(G(x) + G(-x))}_{\text{even}} + \underbrace{\frac{1}{2}(G(x) - G(-x))}_{\text{odd}} \right] \cos\left(\frac{k\pi x}{L}\right) dx \\ &= \frac{1}{2L} \int_{-L}^L G_{\text{even}}(x) \cos\left(\frac{k\pi x}{L}\right) dx \\ &= \frac{1}{L} \int_0^L G_{\text{even}}(x) \cos\left(\frac{k\pi x}{L}\right) dx \\ &= A_k \end{aligned}$$

Recall that in the third line, since an odd function times an even function is odd, the second integral vanishes over the symmetric bounds. This leaves only the even component of $G(x)$.

Question: 8.1.11

Problem 10 tells us that $\mathbf{a}_k = \frac{1}{2}(\mathbf{c}_k + \mathbf{c}_{-k})$ on the usual interval from 0 to π . Find a similar formula for b_k from c_k and c_{-k} . In the reverse direction, find the complex coefficient c_k in $F(x) = \sum c_k e^{ikx}$ from the real coefficients a_k and b_k .

A point of clarification is warranted. Problem 8.1.10 calculates the Fourier coefficient for the *even extension* of $G(x)$. We can do the same for the *odd* extension of $G(x)$, which involves reworking the derivation in the previous problem to ensure that the sine term survives. The way to do so is to begin from $(c_k - c_{-k})/2$. Doing so leaves us with

$$\frac{c_k - c_{-k}}{2} = -\frac{i}{2L} \int_{-L}^L G_{\text{odd}}(x) \sin\left(\frac{k\pi x}{L}\right) dx$$

Multiply by i to conclude

$$i\left(\frac{c_k - c_{-k}}{2}\right) = \frac{1}{2L} \int_{-L}^L G_{\text{odd}}(x) \sin\left(\frac{k\pi x}{L}\right) dx = B_k$$

Now, we consider the original function $G(x)$ (with the definition in Problem 8.1.9). Fix k and write

$$\begin{aligned} c_k &= \frac{1}{2L} \int_{-L}^L G(x) e^{-ik\pi x/L} dx \\ &= \frac{1}{2L} \int_{-L}^L G(x) \left[\cos\left(\frac{k\pi x}{L}\right) - i \sin\left(\frac{k\pi x}{L}\right) \right] dx \\ &= \frac{1}{2}(a_k - ib_k) \end{aligned}$$

Analogously conclude that

$$c_{-k} = \frac{1}{2}(a_k + ib_k)$$

Two equations for two unknowns gives us

$$\begin{aligned} a_k &= c_k + c_{-k} \\ b_k &= i(c_k - c_{-k}) \end{aligned}$$

which are the great and elegant formulae for relating the complex and trigonometric Fourier coefficients for $G(x)$.

Question: 8.1.12

Find the solution to Laplace's equation with $u_0 = \theta$ on the boundary. Why is this the imaginary part of $2(z - z^2/2 + z^3/3 \cdots) = 2\log(1 + z)$? Confirm that on the unit circle $z = e^{i\theta}$, the imaginary part of $2\log(1 + z)$ agrees with θ .

Since $u_0(\theta) = \theta$ is an odd function, we calculate the sine series as

$$\begin{aligned} b_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} \theta \, d\theta \\ &= 0 \\ b_k &= \frac{1}{\pi} \int_{-\pi}^{\pi} \theta \sin(k\theta) \, d\theta \\ &= \frac{1}{\pi} \left[\frac{\sin(k\theta)}{k^2} - \frac{\theta \cos(k\theta)}{k} \right] \Big|_{-\pi}^{\pi} \\ &= \frac{2}{k} (-1)^{k+1} \end{aligned}$$

The solution to Laplace's equation is

$$u(r, \theta) = \frac{2}{1} r \sin(\theta) - \frac{2}{2} r^2 \sin(2\theta) + \frac{2}{3} r^3 \sin(3\theta) - \cdots = \sum_{k=1}^{\infty} \frac{2}{k} (-1)^{k+1} r^k \sin(k\theta)$$

Let $z = re^{i\theta}$. Then this solution indeed corresponds to $\text{Im}[2\log(1 + z)]$.

Question: 8.1.13

If the boundary condition for Laplace's equation is $u_0 = 1$ for $0 < \theta < \pi$ and $u_0 = 0$ for $-\pi < \theta < 0$, find the Fourier series solution $u(r, \theta)$ inside the unit circle. What is u at the origin $r = 0$?

We must calculate the complete Fourier series, which is

$$\begin{aligned} a_0 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} u_0 \, d\theta \\ &= \frac{1}{2\pi} \int_0^{\pi} d\theta \\ &= \frac{1}{2} \\ a_k &= \frac{1}{\pi} \int_0^{\pi} \cos(k\theta) \, d\theta \\ &= 0 \\ b_k &= \frac{1}{\pi} \int_0^{\pi} \sin(k\theta) \, d\theta \\ &= \frac{1}{\pi} \left[\frac{(-1)^{k+1} + 1}{k} \right] \end{aligned}$$

This vanishes for even k , so we should write

$$b_n = \frac{2}{\pi(2n-1)}$$

The solution to Laplace's equation is

$$u(r, \theta) = \frac{1}{2} + \sum_{n=1}^{\infty} \frac{2}{\pi(2n-1)} r^{2n-1} \sin((2n-1)\theta)$$

Question: 8.1.14

With boundary values $u_0(\theta) = 1 + \frac{1}{2}e^{i\theta} + \frac{1}{4}e^{2i\theta} + \dots$, what is the Fourier series solution to Laplace's equation in the circle? Sum this geometric series.

Multiply each term by the corresponding power of r to get

$$u(r, \theta) = \sum_{n=0}^{\infty} \left(\frac{1}{2} r e^{i\theta} \right)^n$$

For this infinite sum, we have closed-form expression

$$u(r, \theta) = \frac{1}{1 - \frac{1}{2} r e^{i\theta}}$$

Question: 8.1.15

- Verify that the fraction in Poisson's formula (30) satisfies Laplace's equation.
- Find the response $u(r, \theta)$ to an impulse at $x = 0$, $y = 1$ (where $\theta = \frac{\pi}{2}$).

- In (r, θ) coordinates, Laplace's equation is

$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0$$

Note that the Poisson's equation has, for the purposes of differentiation, a more legible summation form

$$u(r, \theta) = \frac{1}{2\pi} + \frac{1}{\pi} \sum_{n=1}^{\infty} r^n \cos(n\theta)$$

Each term is calculated as

$$\begin{aligned} \frac{\partial^2 u}{\partial r^2} &= \frac{1}{\pi} \sum_{n=1}^{\infty} n(n-1) r^{n-2} \cos(n\theta) \\ \frac{1}{r} \frac{\partial u}{\partial r} &= \frac{1}{\pi} \sum_{n=1}^{\infty} n r^{n-2} \cos(n\theta) \\ \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} &= -\frac{1}{\pi} \sum_{n=1}^{\infty} n^2 r^{n-2} \cos(n\theta) \end{aligned}$$

which, expectedly, vanish upon summation.

- (b) Think of the impulse as undergoing an angular displacement of $\theta = \pi/2$. The phase shift $\cos(\theta + \pi/2)$ becomes $-\sin(\theta)$. Poisson's equation becomes

$$u(r, \theta) = \frac{1}{2\pi} \frac{1 - r^2}{1 + r^2 + 2r \sin(\theta)}$$

Question: 8.1.16

With complex exponentials in $F(x) = \sum c_k e^{ikx}$, the energy identity (21) changes to $\int_{-\pi}^{\pi} |F(x)|^2 dx = 2\pi \sum |c_k|^2$. Derive this by integrating $(\sum c_k e^{ikx})(\sum \bar{c}_k e^{-ikx})$.

By orthogonality, we know that surviving terms will only be those with the same k in both summations. This leaves us with

$$\int_{-\pi}^{\pi} \sum |c_k|^2 dx = 2\pi \sum |c_k|^2$$

Question: 8.1.17

A centered square wave has $F(x) = 1$ for $|x| \leq \pi/2$.

- Find its energy $\int |F(x)|^2 dx$ by direct integration
- Compute its Fourier coefficients c_k as specific numbers
- Find the sum in the energy identity (Problem 16).

(a)

$$\int_{-\pi}^{\pi} |F(x)|^2 dx = \int_{-\pi/2}^{\pi/2} dx = \pi$$

(b)

$$\begin{aligned} c_k &= \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} e^{-ikx} dx \\ &= -\frac{1}{2\pi ik} \left(e^{-ik\pi/2} - e^{ik\pi/2} \right) \\ &= \frac{1}{\pi k} \sin\left(\frac{k\pi}{2}\right) \end{aligned}$$

In general, the squared coefficients take on form

$$|c_k|^2 = \begin{cases} \frac{1}{4}, & k = 0 \\ \frac{1}{\pi^2 (2k-1)^2}, & k > 0 \end{cases}$$

- (c) Since we need to sum over all integers (positive, negative, and zero), we need to multiply the summation here by two to account for the negatives (as it only sums from $k = 1$ onward), and add $|c_0|^2$.

$$\begin{aligned}
 \int_{-\pi}^{\pi} \sum |c_k|^2 dx &= 2\pi \sum_{-\infty}^{\infty} |c_k|^2 \\
 &= 2\pi \left(|c_0|^2 + \sum_{k=1}^{\infty} (|c_k|^2 + |c_{-k}|^2) \right) \\
 &= 2\pi \left(|c_0|^2 + 2 \sum_{k=1}^{\infty} |c_k|^2 \right) \\
 &= 2\pi \left(\frac{1}{4} + 2 \sum \frac{1}{\pi^2 (2k-1)^2} \right) \\
 &= 2\pi \left(\frac{1}{4} + \frac{1}{4} \right) \\
 &= \pi
 \end{aligned}$$

Question: 8.1.18

$F(x) = 1 + (\cos(x))/2 + \cdots + (\cos(nx))/2^n + \cdots$ is analytic: infinitely smooth.

- (a) If you take 10 derivatives, what is the Fourier series of $d^{10}F/dx^{10}$?
 (b) Does that series still converge quickly? Compare n^{10} with 2^n for $n = 2^{10}$.

- (a) We have

$$F^{10}(x) = - \left[\frac{\cos(x)}{2} + \cdots + \frac{n^{10} \cos(nx)}{2^n} + \cdots \right] = - \sum_{k=1}^{\infty} \frac{k^{10} \cos(kx)}{2^k}$$

which is indeed its own Fourier series.

- (b) Yes; $(2^{10})^{10} = 2^{100}$ is immensely less than $2^{2^{10}}$. In general, n^{10} grows much slower than 2^n for large n .

Question: 8.1.19

If $f(x) = 1$ for $|x| \leq \pi/2$ and $f(x) = 0$ for $\pi/2 < |x| < \pi$, find its cosine coefficients. Can you graph and compute the Gibbs overshoot at the jumps?

The cosine coefficients are

$$a_k = \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} \cos(kx) dx = \frac{2}{\pi k} \sin\left(\frac{k\pi}{2}\right)$$

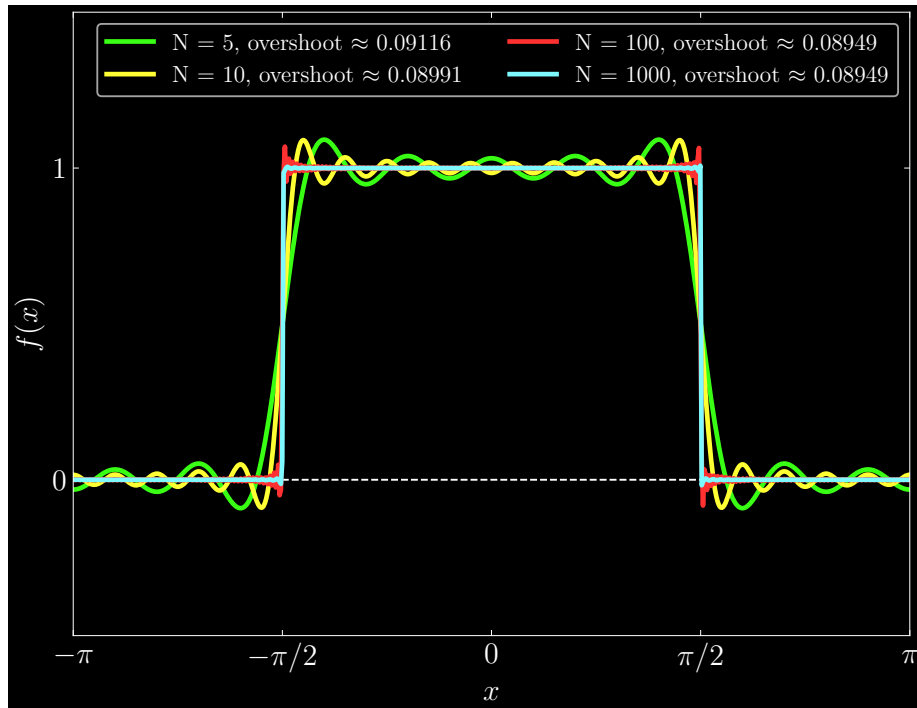
which reduces to

$$a_n = (-1)^{n+1} \frac{2}{\pi(2n-1)}$$

Since $a_0 = 1/2$, the Fourier cosine series is

$$f(x) = \frac{1}{2} + \sum_{n=1}^{\infty} (-1)^{n+1} \frac{2}{\pi(2n-1)} \cos((2n-1)x)$$

The Gibbs overshoot for numerous values of partial sums S_N is



Question: 8.1.20

Find all the coefficients a_k and b_k for F , I , and D on the interval $-\pi \leq x \leq \pi$:

$$F(x) = \delta\left(x - \frac{\pi}{2}\right) \quad I(x) = \int_0^x \delta\left(x - \frac{\pi}{2}\right) dx \quad D(x) = \frac{d}{dx} \delta\left(x - \frac{\pi}{2}\right)$$

$F(x)$:

$$\begin{aligned} a_0 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \delta\left(x - \frac{\pi}{2}\right) dx \\ a_k &= \frac{1}{\pi} \int_{-\pi}^{\pi} \cos(kx) \delta\left(x - \frac{\pi}{2}\right) dx \\ &= \frac{1}{\pi} \cos\left(\frac{k\pi}{2}\right) \\ b_k &= \frac{1}{\pi} \int_{-\pi}^{\pi} \sin(kx) \delta\left(x - \frac{\pi}{2}\right) dx \\ &= \frac{1}{\pi} \sin\left(\frac{k\pi}{2}\right) \end{aligned}$$

$I(x)$: Note that this $I(x)$ is a definition of the Heaviside step function $H\left(x - \frac{\pi}{2}\right)$.

$$\begin{aligned} a_0 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \int_0^x \delta\left(x - \frac{\pi}{2}\right) dx dx \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} H\left(x - \frac{\pi}{2}\right) dx \\ &= \frac{1}{4} \\ a_k &= \frac{1}{\pi} \int_{-\pi}^{\pi} \cos(kx) H\left(x - \frac{\pi}{2}\right) dx \\ &= \frac{1}{\pi k} \left[\sin(k\pi) - \sin\left(\frac{k\pi}{2}\right) \right] \\ &= -\frac{1}{\pi k} \sin\left(\frac{k\pi}{2}\right) \\ b_k &= \frac{1}{\pi} \int_{-\pi}^{\pi} \sin(kx) H\left(x - \frac{\pi}{2}\right) dx \\ &= \frac{1}{\pi k} \left[\cos\left(\frac{k\pi}{2}\right) - \cos(k\pi) \right] \end{aligned}$$

$D(x)$: Relying on the intuition of integration by parts is key, although remember that the delta function is not a “function” in the rigorous sense (for reasons beyond our scope here).

For a_0 , let $u = 1$ and $dv = \frac{d}{dx}\delta\left(x - \frac{\pi}{2}\right) dx$. Then $du = 0$ and $v = \delta\left(x - \frac{\pi}{2}\right)$.

$$\begin{aligned} a_0 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{d}{dx}\delta\left(x - \frac{\pi}{2}\right) dx \\ &= \frac{1}{2\pi} \delta\left(x - \frac{\pi}{2}\right) \Big|_{-\pi}^{\pi} \\ &= 0 \end{aligned}$$

In the cases of a_k and b_k , let $u = \cos(kx)$ or $u = \sin(kx)$, corresponding to $du = -k \sin(kx)$ or $du = k \cos(kx)$.

$$\begin{aligned} a_k &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos(kx) \frac{d}{dx}\delta\left(x - \frac{\pi}{2}\right) dx \\ &= -\frac{k}{2\pi} \int_{-\pi}^{\pi} \sin(kx) \delta\left(x - \frac{\pi}{2}\right) dx \\ &= -\frac{k}{2\pi} \sin\left(\frac{k\pi}{2}\right) \\ b_k &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \sin(kx) \frac{d}{dx}\delta\left(x - \frac{\pi}{2}\right) dx \\ &= \frac{k}{2\pi} \int_{-\pi}^{\pi} \cos(kx) \delta\left(x - \frac{\pi}{2}\right) dx \\ &= \frac{k}{2\pi} \cos(kx) \end{aligned}$$

Observe that a_k and b_k here are the Fourier coefficients of the step function case multiplied by k .

Question: 8.1.21

For the one-sided tall box function in Example 4, with $F = 1/h$ for $0 \leq x \leq h$, what is its odd part $\frac{1}{2}(F(x) - F(-x))$? I am surprised that the Fourier coefficients of this odd part disappear as h approaches zero and $F(x)$ approaches $\delta(x)$.

In order for the even and odd decomposition to work, we should clarify that our F is defined over a symmetric interval $-h \leq x \leq h$, namely

$$F(x) = \begin{cases} 1/h, & 0 \leq x \leq h \\ 0, & -h \leq x < 0 \end{cases}$$

Given the piecewise nature of this function, we must calculate the odd part on each interval. First, over $0 \leq x \leq h$:

$$F_{\text{odd}} = \frac{1}{2} \left(\frac{1}{h} - 0 \right) = \frac{1}{2h}$$

and secondly, over $-h \leq x < 0$:

$$F_{\text{odd}} = \frac{1}{2} \left(0 - \frac{1}{h} \right) = -\frac{1}{2h}$$

Now we calculate the Fourier coefficients

$$\begin{aligned}
 a_0 &= \frac{1}{2h} \int_{-h}^h F(x) \, dx \\
 &= \frac{1}{2h} \left[\int_0^h \frac{1}{2h} \, dx - \int_{-h}^0 \frac{1}{2h} \, dx \right] \\
 &= \frac{1}{2h} \left[\frac{1}{2} - \frac{1}{2} \right] \\
 &= 0 \\
 a_k &= \frac{1}{h} \int_{-h}^h F(x) \cos(kx) \, dx \\
 &= \frac{1}{2h^2} \left[\int_0^h \cos(kx) \, dx - \int_{-h}^0 \cos(kx) \, dx \right] \\
 &= 0 \\
 b_k &= \frac{1}{\pi} \int_{-h}^h F(x) \sin(kx) \, dx \\
 &= \frac{1}{2\pi h} \left[\int_0^h \sin(kx) \, dx - \int_{-h}^0 \sin(kx) \, dx \right] \\
 &= \frac{1}{2\pi kh} [-\cos(kh) + 1 + 1 - \cos(kh)] \\
 &= \frac{\cos(kh) - 1}{\pi kh}
 \end{aligned}$$

In the limit, we can apply l'Hôpital's rule to find that $b_k \rightarrow 0$ as $h \rightarrow 0$, though b_k is never truly zero.

Question: 8.1.22

Find the series $F(x) = \sum c_k e^{ikx}$ for $F(x) = e^x$ on $-\pi \leq x \leq \pi$. That function e^x looks smooth, but there must be a hidden jump to get coefficients c_k proportional to $1/k$. Where is the jump?

The coefficients are

$$\begin{aligned}
 c_k &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^x e^{-ikx} \, dx \\
 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{(1-ik)x} \, dx \\
 &= \frac{1}{2\pi(1-ik)} e^{(1-ik)x} \Big|_{-\pi}^{\pi} \\
 &= \frac{1}{2\pi(1-ik)} [e^{(1-ik)\pi} - e^{-(1-ik)\pi}] \\
 &= \frac{(-1)^k}{2\pi(1-ik)} [e^{\pi} - e^{-\pi}]
 \end{aligned}$$

The jumps occur at every $x = (2k + 1)\pi$, where we go from e^π down to $e^{-\pi}$. That jump appears as a factor in the Fourier coefficients.

Question: 8.1.23

(a) (Old particular solution) Solve $Ay'' + By' + Cy = e^{ikx}$.

(b) (New particular solution) Solve $Ay'' + By' + Cy = \sum c_k e^{ikx}$.

(a) Let $y_p = Y_k e^{ikx}$. Then $y'' = -k^2 Y_k e^{ikx}$ and $y' = ik Y_k e^{ikx}$. We have

$$-k^2 Y_k A + ik Y_k B + Y_k C = 1 \quad \implies \quad Y_k = \frac{1}{C - k^2 A + ikB}$$

(b) By superposition,

$$y_p = \sum c_k Y_k e^{ikx}$$

8.2 - The Fast Fourier Transform

Question: 8.2.1

Multiply the three matrices in equation (11) and compare with F . In which six entries do you need to know that $i^2 = -1$? This is $(w_4)^2 = w_2$. If $M = N/2$, why is $(w_N^M = -1)$?

$$\begin{aligned}
 F_4 &= \begin{bmatrix} 1 & & 1 & \\ & 1 & & i \\ 1 & & -1 & \\ & 1 & & -i \end{bmatrix} \begin{bmatrix} 1 & 1 & & \\ & i^2 & & \\ & & 1 & 1 \\ & & 1 & i^2 \end{bmatrix} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i^2 & i & i^3 \\ 1 & 1 & -1 & -1 \\ 1 & i^2 & -i & -i^3 \end{bmatrix} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & i^2 & i^3 \\ 1 & -1 & 1 & -1 \\ 1 & -i & i^2 & -i^3 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & i^2 & i^3 \\ 1 & i^2 & i^4 & i^6 \\ 1 & i^3 & i^6 & i^9 \end{bmatrix}
 \end{aligned}$$

The right-most three entries of the bottom two rows require knowledge of $i^2 = -1$.

For $M = N/2$, we have

$$w_N^M = w_N^{N/2} = e^{i2\pi \cdot N/2N} = e^{i\pi} = -1$$

These are the roots of unity on opposite ends of the real axis.

Question: 8.2.2

Why is row i of $\overline{F}$ the same as row $N - i$ of F (numbered from 0 to $N - 1$)?

I will switch the index to k as the book's choice is confusing. One intuitive, geometric analogy to make here is $\overline{F}$ *reverses* the direction of the roots of unity starting from zero. As we move down the Fourier matrix row by row, instead of moving counterclockwise around the unit circle to the k -th step, the conjugate changes the sign, causing us to move in the clockwise direction.

The other argument is the powers of $\omega = e^{i2\pi/N}$ are computed modulo N . The k -th row of $\overline{F}$ has $\omega = e^{-i2k\pi/N}$ and the $N - k$ -th row of F has $\omega = e^{i2\pi(N-k)/N} = e^{-i2k\pi/N}$. In this sense, row " N " of F simply corresponds to row 0: which is a row of 1's in both of F and $\overline{F}$.

Question: 8.2.3

From Problem 2, find the 4 by 4 permutation matrix P so that $F = P\bar{F}$. Check that $P^2 = I$ so that $P = P^{-1}$. Then from $\bar{F}F = 4I$ show that $F^2 = 4P$.

It is amazing that $F^4 = 16P^2 = 16I$. *Four transforms of any $\mathbf{c}$ bring back $16\mathbf{c}$.* For all N , F^2/N is a permutation matrix P and $\mathbf{F}^4 = N^2\mathbf{I}$.

In the 4 by 4 case, we need only switch rows 1 and 3:

$$P = \begin{bmatrix} 1 & & & \\ & & 1 & \\ & 1 & & \\ & & & 1 \end{bmatrix}$$

Check that

$$P^2 = \begin{bmatrix} 1 & & & \\ & & 1 & \\ & 1 & & \\ & & & 1 \end{bmatrix} \begin{bmatrix} 1 & & & \\ & & 1 & \\ & 1 & & \\ & & & 1 \end{bmatrix} = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} = I$$

Next, we can prove

$$\begin{aligned} \bar{F}F &= PF^2 \quad (PF = \bar{F} \iff F = P\bar{F}) \\ \implies F^2 &= 4P \end{aligned}$$

Question: 8.2.4

Invert the three factors in equation (11) to find a fast factorization of F^{-1} .

$$\begin{aligned} \begin{bmatrix} 1 & & 1 & \\ & 1 & & i \\ & & -1 & \\ 1 & & & -i \end{bmatrix}^{-1} &= \frac{1}{2} \begin{bmatrix} 1 & & 1 & \\ & 1 & & 1 \\ & & -1 & \\ & -i & & i \end{bmatrix} \\ \begin{bmatrix} 1 & 1 & & \\ 1 & i^2 & & \\ & & 1 & 1 \\ & & 1 & i^2 \end{bmatrix}^{-1} &= \frac{1}{2} \begin{bmatrix} 1 & 1 & & \\ 1 & -1 & & \\ & & 1 & 1 \\ & & 1 & -1 \end{bmatrix} \\ \begin{bmatrix} 1 & & & \\ & & 1 & \\ & 1 & & \\ & & & 1 \end{bmatrix}^{-1} &= \begin{bmatrix} 1 & & & \\ & & 1 & \\ & 1 & & \\ & & & 1 \end{bmatrix} \end{aligned}$$

The fast factorization of F_4^{-1} is

$$F_4^{-1} = \frac{1}{4} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & & \\ 1 & -1 & & \\ & & 1 & 1 \\ & & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & & 1 & \\ & 1 & & 1 \\ & & -1 & \\ & -i & & i \end{bmatrix}$$

Question: 8.2.5

F is symmetric. Transpose equation (11) to find a new Fast Fourier Transform.

$$F^T = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & & \\ 1 & i^2 & & \\ & & 1 & 1 \\ & & 1 & i^2 \end{bmatrix} \begin{bmatrix} 1 & & 1 & \\ & 1 & & 1 \\ 1 & & -1 & \\ & i & & -i \end{bmatrix}$$

Question: 8.2.6

All entries in the factorization of F_6 involve powers of $w =$ sixth root of 1:

$$F_6 = \begin{bmatrix} I & D \\ I & -D \end{bmatrix} \begin{bmatrix} F_3 & \\ & F_3 \end{bmatrix} \begin{bmatrix} & P \\ & \end{bmatrix}$$

Write down these factors with 1, w , w^2 in D and powers of w^2 in F_3 . Multiply!

$$I = \begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & & \\ & w & \\ & & w^2 \end{bmatrix}, \quad F_3 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & w^2 & (w^2)^2 \\ 1 & (w^2)^2 & (w^2)^4 \end{bmatrix}$$

$$P = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix}$$

A point of confusion is in conflating w_3 and w_6 , so to clarify, above we have $w = w_6$. Use the relationship $(w_6)^2 = w_3$. This gives us F_3 . Multiplying the factors gives us

$$\begin{aligned}
F_6 &= \begin{bmatrix} 1 & & & & & \\ & 1 & & & & \\ & & 1 & & & \\ & & & -1 & & \\ & & & & -w & \\ & & & & & -w^2 \end{bmatrix} \begin{bmatrix} 1 & & & & & \\ & 1 & & & & \\ & & w^2 & & & \\ & & & (w^2)^2 & & \\ & & & & (w^2)^4 & \\ & & & & & 1 \end{bmatrix} \begin{bmatrix} 1 & & & & & \\ & & 1 & & & \\ & & & 1 & & \\ & & & & 1 & \\ & & & & & 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 & & & & & \\ & 1 & & & & \\ & & 1 & & & \\ & & & -1 & & \\ & & & & -w & \\ & & & & & -w^2 \end{bmatrix} \begin{bmatrix} 1 & & & & & \\ & 1 & & & & \\ & & w^2 & & & \\ & & & (w^2)^2 & & \\ & & & & (w^2)^4 & \\ & & & & & 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & w & w^2 & w^3 & (w^2)^2 & w^5 \\ 1 & w^2 & (w^2)^2 & (w^2)^3 & (w^2)^4 & (w^2)^5 \\ 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & -w & w^2 & -w^3 & (w^2)^2 & -w^5 \\ 1 & -w^2 & (w^2)^2 & -(w^2)^3 & (w^2)^2 & -(w^2)^5 \end{bmatrix}
\end{aligned}$$

Question: 8.2.7

Put the vector $\mathbf{c} = (1, 0, 1, 0)$ through the three steps of the FFT to find $\mathbf{y} = F\mathbf{c}$. Do the same for $\mathbf{c} = (0, 1, 0, 1)$.

$$\begin{aligned}
F_4 c_1 &= \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -i \end{bmatrix} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & i^2 \end{bmatrix} \begin{bmatrix} 1 & & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \\
&= \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -i \end{bmatrix} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & i^2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -i \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} \\
&= \begin{bmatrix} 2 \\ 0 \\ 2 \\ 0 \end{bmatrix}
\end{aligned}$$

$$\begin{aligned}
F_4 c_2 &= \begin{bmatrix} 1 & 1 & & \\ & 1 & i & \\ 1 & & -1 & \\ & 1 & & -i \end{bmatrix} \begin{bmatrix} 1 & 1 & & \\ & 1 & i^2 & \\ & & 1 & 1 \\ & & 1 & i^2 \end{bmatrix} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 1 & & \\ & 1 & i & \\ 1 & & -1 & \\ & 1 & & -i \end{bmatrix} \begin{bmatrix} 1 & 1 & & \\ & 1 & i^2 & \\ & & 1 & 1 \\ & & 1 & i^2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 1 & & \\ & 1 & i & \\ 1 & & -1 & \\ & 1 & & -i \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix} \\
&= \begin{bmatrix} 2 \\ 0 \\ -2 \\ 0 \end{bmatrix}
\end{aligned}$$

Question: 8.2.8

Compute $\mathbf{y} = F_8 \mathbf{c}$ by the three FFT steps for $\mathbf{c} = (1, 0, 1, 0, 1, 0, 1, 0)$. Repeat the computation for $\mathbf{c} = (0, 1, 0, 1, 0, 1, 0, 1)$.

Let $w = w_8$. We have

$$\begin{aligned}
F_8 &= \begin{bmatrix} I_4 & D_4 \\ I_4 & -D_4 \end{bmatrix} \begin{bmatrix} F_4 & \\ & F_4 \end{bmatrix} \begin{bmatrix} & P_8 \\ & \end{bmatrix} \\
I_4 &= \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix}, \quad D_4 = \begin{bmatrix} 1 & & & \\ & w^2 & & \\ & & (w^2)^2 & \\ & & & (w^2)^4 \end{bmatrix}
\end{aligned}$$

where we use the fact that $(w_8)^2 = w_4$. The permutation matrix is

$$P_8 = \begin{bmatrix} 1 & & & & & & & \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & 1 & & & & \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & 1 & & & & \\ & & & & 1 & & & \\ & & & & & 1 & & \\ & & & & & & 1 & \end{bmatrix}$$

Through recursion, and further exploiting that $(w_4)^2 = w_2 = w^4$, we further decompose into

$$\begin{bmatrix} F_4 & \\ & F_4 \end{bmatrix} = \begin{bmatrix} I_2 & D_2 & & \\ I_2 & -D_2 & & \\ & & I_2 & D_2 \\ & & I_2 & -D_2 \end{bmatrix} \begin{bmatrix} F_2 & & & \\ & F_2 & & \\ & & F_2 & \\ & & & F_2 \end{bmatrix} \begin{bmatrix} \text{pick} & 0, 4 \\ \text{pick} & 2, 6 \\ \text{pick} & 1, 5 \\ \text{pick} & 3, 7 \end{bmatrix}$$

$$I_2 = \begin{bmatrix} 1 & \\ & 1 \end{bmatrix}, \quad D_2 = \begin{bmatrix} 1 & \\ & w^4 \end{bmatrix}, \quad F_2 = \begin{bmatrix} 1 & 1 \\ 1 & w^4 \end{bmatrix}$$

Computing, $\mathbf{c}_1$ outputs $(4, 0, 0, 0, 4, 0, 0, 0)$ and $\mathbf{c}_2$ gives $(4, 0, 0, 0, -4, 0, 0, 0)$.

Question: 8.2.9

If $w = e^{2\pi i/64}$ then w^2 and $\sqrt{w}$ are among the ___ and ___ roots of 1.

We have w^2 is a primitive 32^{th} root of unity and $\sqrt{w}$ is a primitive 128^{th} root of unity. Both generate the other roots of unity for their respective N .

Question: 8.2.10

F is a symmetric matrix. Its eigenvalues aren't real. How is this possible?

The theorem requires that the symmetric matrix itself must be real to have real eigenvalues. By contrast, F definitionally has imaginary components in its entries, so that assumption is not satisfied.

Question: 8.2.11

The eigenvectors of K_N and B_N are the discrete sines $\mathbf{s}_1, \dots, \mathbf{s}_N$ and the discrete cosines $\mathbf{c}_0, \dots, \mathbf{c}_{N-1}$. Notice the eigenvector $\mathbf{c}_0 = (1, 1, \dots, 1)$. Here are the $\mathbf{s}_k$ and $\mathbf{c}_k$ —these vectors are samples of $\sin(kx)$ and $\cos(kx)$ from 0 to π .

$$\left(\sin\left(\frac{\pi k}{N+1}\right), \sin\left(\frac{2\pi k}{N+1}\right), \dots, \sin\left(\frac{N\pi k}{N+1}\right) \right)$$

and

$$\left(\cos\left(\frac{\pi k}{2N}\right), \cos\left(\frac{3\pi k}{2N}\right), \dots, \cos\left(\frac{(2N-1)\pi k}{2N}\right) \right)$$

For 2 by 2 matrices K_2 and B_2 , verify that $\mathbf{s}_1, \mathbf{s}_2$ and $\mathbf{c}_0, \mathbf{c}_1$ are eigenvectors.

See the inset on the symmetric tridiagonal matrices in the textbook. In the cases of K_2 and B_2 , we have

$$K_2 = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, \quad B_2 = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

The eigenvectors are

$$\mathbf{s}_1 = \begin{bmatrix} \sqrt{3}/2 \\ \sqrt{3}/2 \end{bmatrix}, \quad \mathbf{s}_2 = \begin{bmatrix} \sqrt{3}/2 \\ -\sqrt{3}/2 \end{bmatrix}$$

$$\mathbf{c}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{c}_1 = \begin{bmatrix} \sqrt{2}/2 \\ -\sqrt{2}/2 \end{bmatrix}$$

Multiplying, we find

$$\begin{aligned} K_2 \mathbf{s}_1 &= \mathbf{s}_1, & K_2 \mathbf{s}_2 &= 3\mathbf{s}_2 \\ B_2 \mathbf{c}_0 &= \mathbf{0}, & B_2 \mathbf{c}_1 &= 2\mathbf{c}_1 \end{aligned}$$

Question: 8.2.12

Show that C_3 has eigenvalues $\lambda = 0, 3, 3$ with eigenvectors $\mathbf{e}_0 = (1, 1, 1)$, $\mathbf{e}_1 = (1, w, w^2)$, $\mathbf{e}_2 = (1, w^2, w^4)$. You may prefer the real eigenvectors $(1, 1, 1)$ and $(1, 0, -1)$ and $(1, -2, 1)$.

We have

$$C_3 = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$$

Now, let $w = w_3$ be a third root of unity. This is easiest if we rely on our geometric intuition. Envision the roots of unity as complex vectors tracing out the unit circle. Treat each equation as a vector addition problem. Doing so allows us to easily calculate

$$\begin{aligned} C_3 \mathbf{e}_0 &= \mathbf{0} \\ C_3 \mathbf{e}_1 &= \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ w \\ w^2 \end{bmatrix} = \begin{bmatrix} 2 - w - w^2 \\ -1 + 2w - w^2 \\ -1 - w + 2w^2 \end{bmatrix} = 3\mathbf{e}_1 \\ C_3 \mathbf{e}_2 &= \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ w^2 \\ w^4 \end{bmatrix} = \begin{bmatrix} 2 - w^2 - w^4 \\ -1 + 2w^2 - w^4 \\ -1 - w^2 + 2w^4 \end{bmatrix} = 3\mathbf{e}_2 \end{aligned}$$

Question: 8.2.13

Multiply to see the eigenvectors $\mathbf{e}_k$ and eigenvalues λ_k of C_N . Simplify to $\lambda_k = 2 - 2\cos(2\pi k/N)$. Explain why C_N is only semidefinite. It is not positive definite.

$$C\mathbf{e}_k = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ w^k \\ w^{2k} \\ w^{(N-1)k} \end{bmatrix} = (2 - w^k - w^{-k}) \begin{bmatrix} 1 \\ w^k \\ w^{2k} \\ w^{(N-1)k} \end{bmatrix}$$

We can derive

$$\lambda_k = 2 - w^k - w^{-k} = 2 - (e^{i2\pi k/N} + e^{-i2\pi k/N}) = 2 - 2\cos(2\pi k/N)$$

Immediately, we can conclude that $\lambda_k \geq 0$ for all k , as the cosine term is bounded by 1. We have $\lambda_k = 0$ only when k is a multiple of N . Equivalently, C_N is semidefinite.

Question: 8.2.14

The eigenvectors $\mathbf{e}_k$ of C are automatically perpendicular because C is a ___ matrix. (To tell the truth, C has repeated eigenvalues as in Problem 12. There was a plane of eigenvectors for $\lambda = 3$ and we chose orthogonal $\mathbf{e}_1$ and $\mathbf{e}_2$ in that plane.)

C is a **symmetric** matrix, which always has orthogonal eigenvectors.

Question: 8.2.15

Write the 2 eigenvalues for K_2 and the 3 eigenvalues for B_3 . Always K_N and B_{N+1} have the same N eigenvalues, with the extra eigenvalue ___ for B_{N+1} . (This is because $K = A^T A$ and $B = A A^T$.)

The characteristic polynomials are

$$\begin{aligned}
 K_2 &= \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \\
 \implies (2 - \lambda)^2 - 1 &= 0 \\
 \implies \lambda_{1,2} &= 1, 3 \\
 B_3 &= \begin{bmatrix} 1 & -1 & \\ -1 & 2 & -1 \\ & -1 & 1 \end{bmatrix} \\
 \implies (1 - \lambda)[(2 - \lambda)(1 - \lambda) - 1] - (1 - \lambda) &= 0 \\
 = (1 - \lambda)[(2 - \lambda)(1 - \lambda) - 2] &= 0 \\
 \implies \lambda_{1,2,3} &= 0, 1, 3
 \end{aligned}$$

The extra eigenvalue of B_3 is $\lambda = 0$.

8.3 - The Heat Equation

Question: 8.3.1

Solve the heat equation $u_t = cu_{xx}$ on an infinite bar with coefficient c , starting from $u = e^{ikx}$ at $t = 0$. As in (10) the solution has the product form $u = e^{i\omega t} e^{ikx}$. With c in the equation, find ω for each k .

With the given solution form, write

$$u_t = (i\omega) e^{i\omega t} e^{ikx} = c(ik)^2 e^{i\omega t} e^{ikx} = u_{xx}$$

Setting the coefficients equal gives us

$$i\omega = c(ik)^2 \implies i\omega = -ck^2$$

Question: 8.3.2

Solve the same equation $u_t = cu_{xx}$ starting from the point source $u = \delta(x) = \int e^{ikx} dk / 2\pi$ at $t = 0$. By superposition, you integrate over all k the solutions u in Problem 1. The result is the heat kernel as in equation (14) but adjusted for c .

By superposition, the solution with the given initial condition is

$$U(t, x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ck^2 t} e^{ikx} dk$$

To solve, differentiate by x :

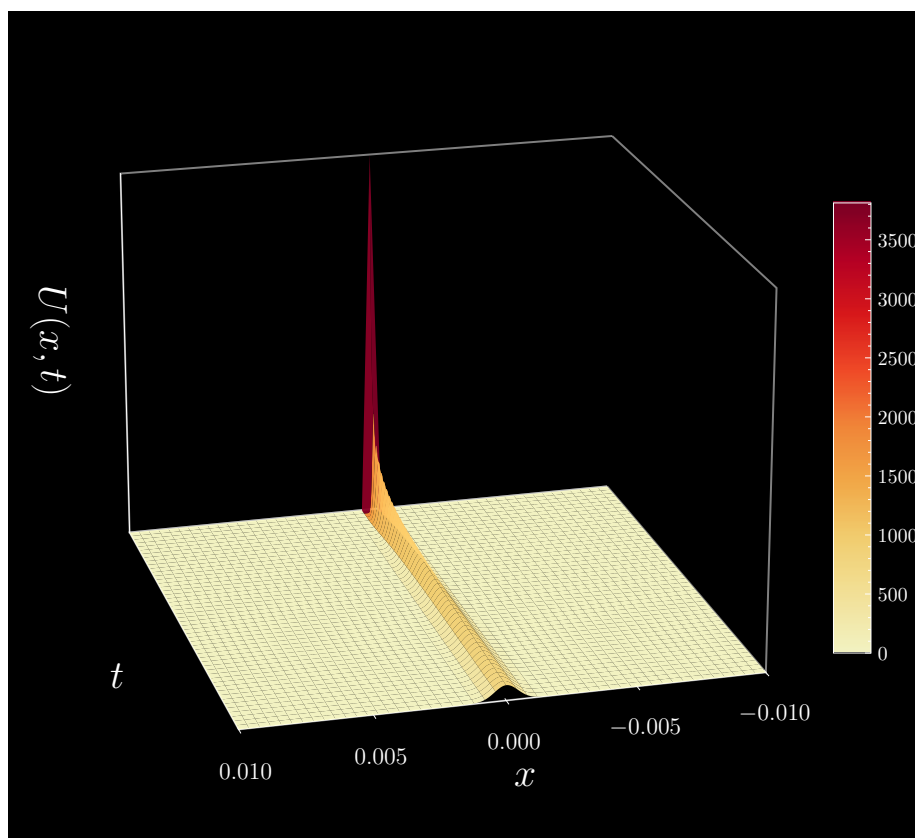
$$\frac{d}{dx} U(t, x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ck^2 t} (ik) e^{ikx} dk$$

Proceed by integration by parts. Let $a = ie^{ikx}$ and $db = ke^{-ck^2 t} dk$. Then $da = -xe^{ikx} dk$ and $b = -(1/2ct) e^{-ck^2 t}$, and we have

$$\begin{aligned} \frac{dU}{dx} &= \frac{1}{2\pi} \left(-\frac{i}{2} cte^{ikx} e^{-ck^2 t} \Big|_{-\infty}^{\infty} - \frac{x}{2ct} \int_{-\infty}^{\infty} e^{-ck^2 t} e^{ikx} dk \right) \\ &= -\frac{x}{2ct} U \\ \implies \frac{dU}{U} &= -\frac{x}{2ct} dx \\ \implies U &= Me^{-x^2/4ct} \end{aligned}$$

Assume a heat kernel of $\int \delta(x) dx = 1$. To normalize this Gaussian integral, set $M = 1/\sqrt{4\pi ct}$. Then our solution is

$$U = \frac{1}{\sqrt{4\pi ct}} e^{-x^2/4ct}$$

**Question: 8.3.3**

To solve $u_t = cu_{xx}$ for a bar between $x = 0$ and $x = 1$, the basis functions are still $\sin(k\pi x)$ (with $u = 0$ at the ends). What are the eigenvalues λ_k that go into the solution $\sum b_k(0) e^{-\lambda_k t} \sin(k\pi x)$?

With the diffusion constant, we now have

$$\lambda_k = ck^2\pi^2$$

On the left, $\partial u / \partial t$ gives us $ck^2\pi^2$. On the right, $\partial^2 u / \partial x^2$ gives us $k^2\pi^2$. Multiply by the diffusivity constant for equality.

Question: 8.3.4

Following Problem 3, solve $u_t = cu_{xx}$ when the initial temperature is $u_0 = 1$ for $\frac{1}{4} \leq x \leq \frac{3}{4}$ (and $u_0 = 0$ on the first and last quarters of the bar). The problem is to find the coefficients $b_k(0)$ for that initial temperature.

First, derive the Fourier series for the even square wave

$$SW(x) = \begin{cases} 1 & \text{for } -\frac{3}{4} < x < -\frac{1}{4} \\ 0 & \text{for } -1 < x < -\frac{3}{4} \text{ and } -\frac{1}{4} < x < \frac{1}{4} \text{ and } \frac{3}{4} < x < 1 \\ 1 & \text{for } \frac{1}{4} < x < \frac{3}{4} \end{cases}$$

An intuitive approach here is to alter the Fourier series for the square wave given in the book's equation (6):

$$\frac{4}{\pi} \sum_{k=1}^{\infty} \frac{\sin[(2k-1)\pi x]}{2k-1}$$

First is to offset the horizontal position by $3/4$ and adjust the periodicity by a factor of 2: instead of the width of the period being 1, it is now $1/2$.

$$SW(x) = \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{\sin[(4k-2)\pi(x + \frac{3}{4})]}{2k-1}$$

By the angle difference identity for sine and reducing trigonometric functions to even-odd sign parity, this becomes

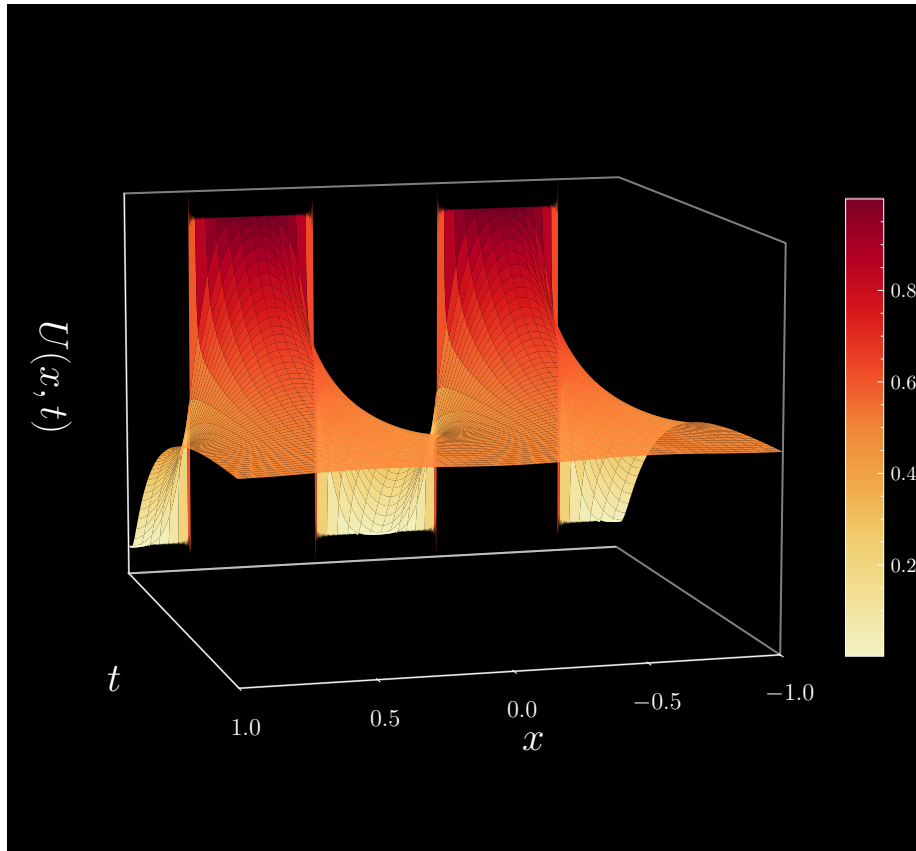
$$SW(x) = \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{(-1)^k}{2k-1} \cos[(4k-2)\pi x]$$

Lastly, we need to scale the amplitude and vertically offset the square wave. For this problem, multiply the sum by $1/2$ (since the wave peaks between 0 and 1) and add $1/2$ (since it is symmetric about $u = 1/2$). This gives us our transformed square wave.

$$SW(x) = \frac{1}{2} + \frac{2}{\pi} \sum_{k=1}^{\infty} \frac{(-1)^k}{2k-1} \cos[(4k-2)\pi x]$$

The corresponding coefficients are then

$$b_k(0) = \frac{2}{\pi} \frac{(-1)^k}{2k-1}$$

**Question: 8.3.5**

Solve the heat equation $u_t = u_{xx}$ from a point source $u(x, 0) = \delta(x)$ with free boundary conditions $u'(\pi, t) = u'(-\pi, t) = 0$. Use the infinite cosine series $\delta(x) = (1 + 2 \cos(x) + 2 \cos(2x) + \dots) / 2\pi$ multiplied by time decay factors $b_k(t)$.

Incorporating the time decay factors, the solution is

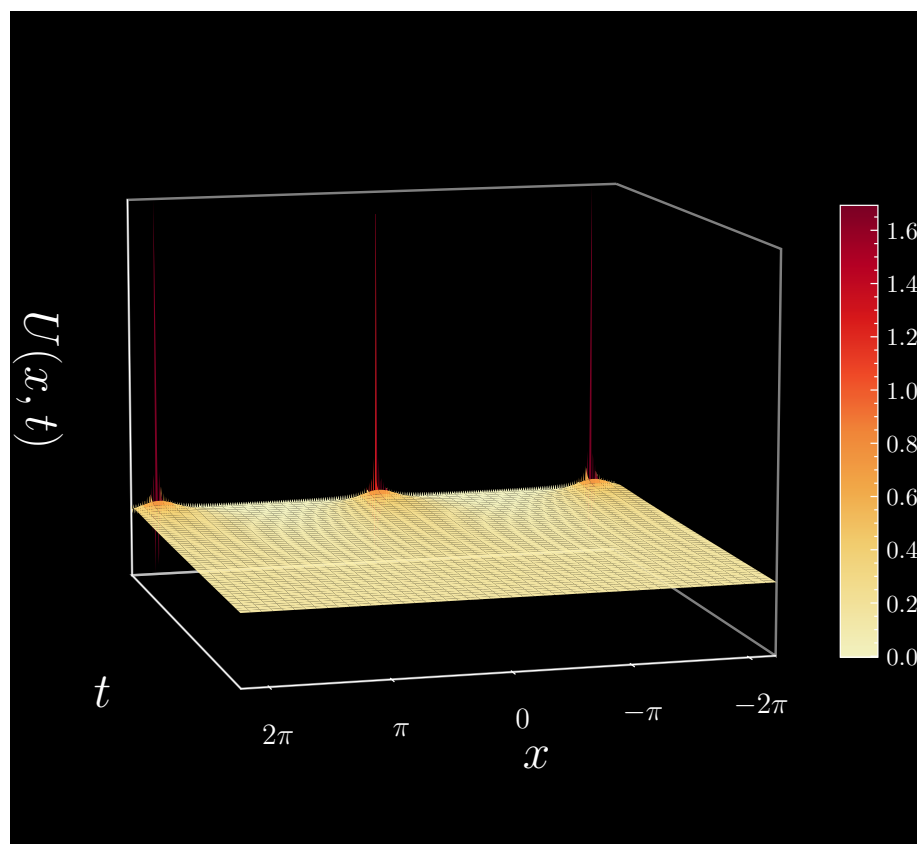
$$u(x, t) = \frac{1}{2\pi} + \frac{1}{\pi} \sum_{k=1}^{\infty} e^{-k^2 \pi^2 t} \cos(kx)$$

Note that the Fourier coefficients to the cosines series provide us with the $b_k(0)$ constants. Differentiating with respect to x , we get

$$u'(x, t) = -\frac{1}{\pi} \sum_{k=1}^{\infty} e^{-k^2 \pi^2 t} k \sin(kx)$$

From which it is immediately apparent that

$$u'(\pi, t) = u'(-\pi, t) = 0$$

**Question: 8.3.6**

(Bar from $x = 0$ to $x = \infty$) Solve $u_t = u_{xx}$ on the positive half of an infinite bar, starting from the shifted delta function $u_0 = \delta(x - a)$ at a point $x = a > 0$. Here is a way to use the full-bar heat kernel U in (14), and still keep $u = 0$ at $x = 0$.

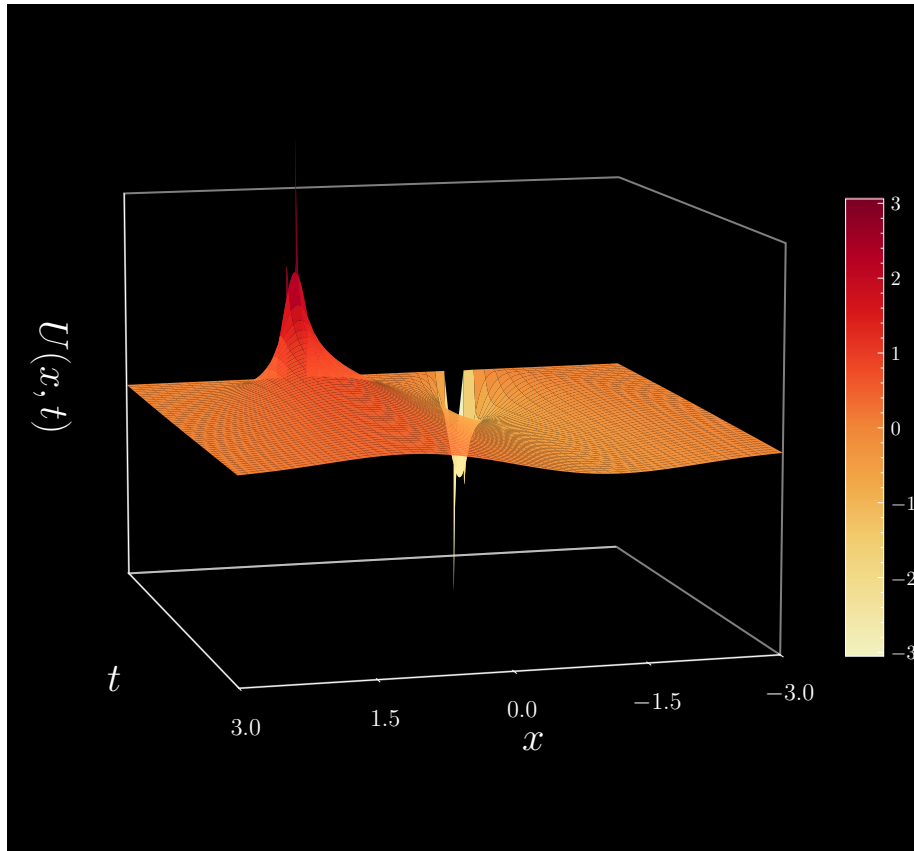
Imagine a negative point source at $x = -a$. Solve the heat equation on the fully infinite bar, including both sources in $u_0 = \delta(x - a) - \delta(x + a)$ at $t = 0$. Your solution (a difference of heat kernels) will stay zero at the boundary $x = 0$ (*Why?*). Then it must be the correct solution on the half-bar, since it started correctly.

Using equation (14) and invoking superposition, our solution is

$$u(t, x) = \frac{1}{\sqrt{4\pi t}} \left[e^{-(x-a)^2/4t} - e^{-(x+a)^2/4t} \right]$$

Note that at $x = 0$, $u(t, 0) = 0$. This method of solution is called the **method of images**. It is called so because of the trick in dropping in another source that generates the desired boundary condition.

For $a = 1$, we can graph the solution as

**Question: 8.3.7**

Check that the basis functions $s_k = \sin\left(k + \frac{1}{2}\right) \pi x$ are orthogonal over $0 \leq x \leq 1$. Find a formula for the coefficient B_4 in the Fourier series $F(x) = \sum B_k s_k$. (Multiply by $s_4(x)$ and integrate, to isolate B_4 .)

Let $k_1 \neq k_2$ be integers. Then we can ascertain

$$\begin{aligned} \int_0^1 \sin\left(k_1 + \frac{1}{2}\right) \pi x \sin\left(k_2 + \frac{1}{2}\right) \pi x \, dx &= \frac{1}{2} \int_0^1 [\cos(k_1 - k_2) \pi x - \cos(k_1 + k_2 + 1) \pi x] \, dx \\ &= \frac{1}{2\pi} \left[\frac{\sin(k_1 - k_2) \pi}{k_1 - k_2} - \frac{\sin(k_1 + k_2 + 1) \pi}{k_1 + k_2 + 1} \right] \\ &= 0 \end{aligned}$$

This is true for any choice of k_1, k_2 . Invoking orthogonality, we can then calcu-

late

$$\begin{aligned}\int_0^1 F(x) s_4(x) dx &= \int_0^1 \sum B_k s_k s_4 dx \\ &= B_4 \int_0^1 \sin^2\left(\frac{9\pi x}{2}\right) dx \\ &= \frac{B_4}{2}\end{aligned}$$

Leaving us with

$$B_4 = 2 \int_0^1 F(x) s_4(x) dx$$

Question: 8.3.8

The basis functions $\sin(k + \frac{1}{2})\pi x$ are for **fixed-free** boundaries ($u = 0$ at $x = 0$ and $u' = 0$ at $x = 1$). What are the basis functions for **free-fixed boundaries** ($u' = 0$ at $x = 0$ and $u = 0$ at $x = 1$)?

For free-fixed boundaries, our basis functions are $\cos(k + \frac{1}{2})\pi x$. Easy to verify that the free-fixed boundary conditions are met.

Question: 8.3.9

Suppose $\mathbf{u}_t = \mathbf{u}_{xx} - \mathbf{u}$ with boundary condition $u = 0$ at $x = 0$ and $x = 1$. Find the new numbers λ_k in the general solution $u = \sum b_k(0) e^{-\lambda_k t} \sin(k\pi x)$. (Previously $\lambda_k = -k^2\pi^2$, now there is a new term in λ because of $-u$.)

The additional $-u$ term adds one to the eigenvalue after canceling the negative, giving us $\lambda_k = k^2\pi^2 + 1$.

Question: 8.3.10

Explain each step in equation (13). Solve $dU/dx = -xU/2t$ to reach $U = e^{-x^2/4t}$. How do the known infinite integrals $\int e^{-x^2} dx = \sqrt{\pi}$ and $\int u dx = 1$ lead to the factor $1/\sqrt{4\pi t}$?

The integration by parts proceeds as follows: let $a = ie^{ikx}$ and $db = e^{-k^2 t} k dk$. Then $da = -xe^{ikx} dk$ and $b = -e^{-k^2 t}/2t$. Combining the parts (don't forget the $1/2\pi$ term) gives

$$\begin{aligned}\frac{dU}{dx} &= -ie^{-k^2 t} e^{ikx} \Big|_{-\infty}^{\infty} - \frac{1}{4\pi t} \int_{-\infty}^{\infty} (e^{-k^2 t}) (xe^{ikx}) dk \\ &= -\frac{1}{4\pi t} \int_{-\infty}^{\infty} (e^{-k^2 t}) (xe^{ikx}) dk \\ &= -\frac{xU}{2t}\end{aligned}$$

Separating variables, we can deduce from $dU/dx = -xU/2t$ that

$$\int \frac{dU}{U} = - \int \frac{x}{2t} dx \implies \ln(U) = -\frac{x^2}{2t} + C \implies U = ce^{-x^2/2t}$$

The normalization constant c follows from the requirement that

$$\int_{-\infty}^{\infty} ce^{-x^2/2t} dx = 1$$

which is true only when $c = 1/\sqrt{4\pi t}$.

Question: 8.3.11

(Shift invariance) What is the solution to $u_t = u_{xx}$ starting from $\delta(x-a)$ at $t=0$?

Equation (18): an integral of shifted heat kernels.

$$u(t, x) = \frac{1}{\sqrt{4\pi t}} \int_{-\infty}^{\infty} u_0(a) e^{-(x-a)^2/4t} da$$

Question: 8.3.12

What are basis functions $A(x, y)$ for heat flow in a square plate, when $u = 0$ along the four sides $x = 0, x = 1, y = 0, y = 1$? The heat equation is $u_t = u_{xx} + u_{yy}$. Find eigenfunctions for $A_{xx} + A_{yy} = \lambda A$ that satisfy the boundary conditions.

The first eigenfunction is $A_{11} = \sin(\pi x) \sin(\pi y)$. Find the eigenvalues λ .

For heat flowing out of the perimeter of the square, we have basis functions

$$A(x, y) = \sin(m\pi x) \sin(n\pi y)$$

The eigenfunctions and corresponding eigenvalues are

$$\begin{aligned} -\pi^2(m^2 + n^2) \sin(m\pi x) \sin(n\pi y) &= \lambda_{m,n} \sin(m\pi x) \sin(n\pi y) \\ \implies \lambda_{m,n} &= -\pi^2(m^2 + n^2) \end{aligned}$$

Question: 8.3.13

Substitute $U = e^{-x^2/4t}/\sqrt{4\pi t}$ to show that this heat kernel solves $U_t = U_{xx}$.

Calculate

$$U_t = \left[\frac{x^2}{4t^2} - \frac{1}{2t} \right] \frac{e^{-x^2/4t}}{\sqrt{4\pi t}} = U_{xx}$$

8.4 - The Wave Equation